Values of $L(x)$ are estimated using its relationship with $l(x)$ and $d(x)$ as follows:

$$L(4) = \frac{l(4)+l(5)}{2}$$

$$= \frac{95000+94500}{2}$$

$$= 94{,}750$$

$$L(5) = l(5) - \frac{d(5)}{2}$$

$$= 94500 - \frac{400}{2}$$

$$= 94{,}300.$$

The value of $T(5)$ is estimated from the given value of $T(4)$ and the estimate of $L(4)$ as

$T(5) = T(4) - L(4)$

$T(5) = 4850300 - 94750 = 47{,}55{,}550.$

The life expectancy of the ***cohort*** at the age $x = 4$ and 5 years is estimated using the relationship

$$e^{\circ}(x) = \frac{T(x)}{l(x)}$$

as

$$e^{\circ}(4) = \frac{4850300}{95000} = 51.06$$

$$e^{\circ}(5) = \frac{4755550}{94500} = 50.32$$

The completed Life Table is

x	$l(x)$	$d(x)$	$p(x)$	$q(x)$	$L(x)$	$T(x)$	$e^0(x)$
4	95,000	500	0.995	0.005	94,750	48,50,300	51.06
5	94,500	400	0.996	0.004	94,300	47,55,550	50.32

8.1.5 Fertility and its Measurements

Fertility refers to births occurring to the women who are at child bearing age. A woman at child bearing age is defined as the age of the female who can give birth to a child. In other words, it is the reproductive age of woman.

Fertility rates are the quantitative characteristics, which are used to measure the rate of growth of the population due to births during a specified period, usually a year. The fertility rates are expressed *per thousand* women who are at child bearing age.

As like mortality rates, there are several fertility rates. Among them, the following are the basic fertility rates discussed here

(i) Crude Birth Rate

(ii) Specific Fertility Rate

(iii) General Fertility Rate

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